BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Final GCSE Computer Science Consolidation

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What should guide the choice of a final revision topic?

2. How many colours can a bitmap with 4 bits per pixel use?

A) 4 B) 8
C) 32 D) 16

3. What happens to raw size when both bitmap dimensions double at fixed depth?

4. A microphone records digital sound. "Sampling stores every instant of the sound wave." Correct this. How does it affect recorded detail?

Today's objectives

Explain the core ideas of final gcse computer science consolidation using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Recall the full course

Paper 1 systems and Paper 2 computational thinking are equal components.

Use accurate language

State precise definitions and causal effects.

Demonstrate independence

Complete unseen traces, calculations and explanation without hints.

Check answers

Use command words, units, bounds and allocated marks before submission.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Recall the full course	Paper 1 systems and Paper 2 computational thinking are equal components.
Use accurate language	State precise definitions and causal effects.
Demonstrate independence	Complete unseen traces, calculations and explanation without hints.
Check answers	Use command words, units, bounds and allocated marks before submission.

I Do - One worked example

Solve an integrated question: a 20 x 20 bitmap with 8-bit depth is uploaded over a network. Calculate uncompressed bytes and explain one network factor.

1. Pixels =20 x 20=400.
2. Bits =400 x 8=3200.
3. Bytes=3200 / 8=400.
4. Bandwidth, latency or congestion can affect upload time; explain the chosen factor.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A list [2,6,10,14,18] is sorted. Trace binary search for 18.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Print the combined size of the first and last file in this list.

```
array filesMB = [2,6,10]
print(...)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Give one difference between compiler and interpreter. [3 marks]

Q3. Calculate 10101010 binary in denary. [2 marks]

Q4. Explain how to make a program robust against invalid numeric input. [4 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. OCR J277 has two equally weighted examination components. ☐ T ☐ F
2. Paper 1 systems and Paper 2 computational thinking are equal components. ☐ T ☐ F
3. State precise definitions and causal effects. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Evidence of a specific knowledge or method gap from the recent targeted review. ☐

Q2 D - 16 ☐

Q3 The number of pixels and raw pixel-data size become four times as large. ☐

Q4 Measure amplitude at regular intervals. Store each rounded (quantised) value in binary. Sample rate and bit depth limit the detail captured. ☐

Code First - Model

```
array filesMB = [2,6,10]
print(filesMB[0] + filesMB[2])
```

Output: 12

We Do - Check the model

1. Found at index4. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Compiled source is translated to executable/intermediate output before execution of translated program; interpreted execution is mediated by an interpreter. ☐

Q3 170. ☐

Q4 Use type/range checks; handle failure safely and request another value; test boundary/erroneous cases. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: